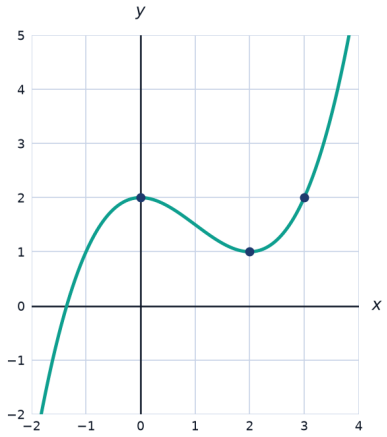
Transformations & Vectors

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Solving equations with algebra

For the model $f(x) = 2x + 9$ in solving equations with algebra, evaluate $f(5)$.

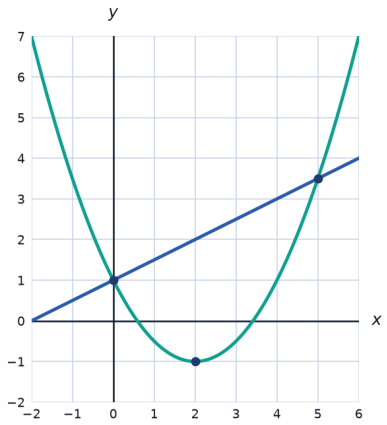


2 [Foundational] • Target: Solving equations with algebra

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

3 [Foundational] • Target: Methods for solving two-step equations

For the model $f(x) = 3x + 3$ in methods for solving two-step equations, evaluate $f(2)$.



4 [Foundational] • Target: Methods for solving two-step equations

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

5 [Foundational] • Target: Methods for solving two-step equations

A data set recorded for methods for solving two-step equations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

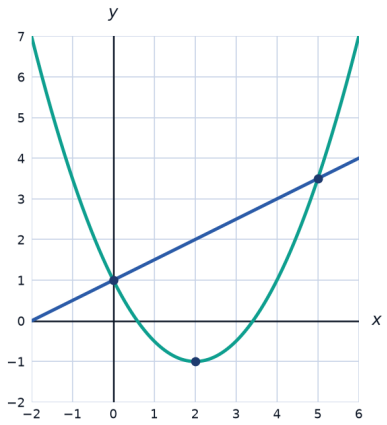
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6 [Foundational] • Target: Methods for solving two-step equations

For the model $f(x) = 3x + 3$ in methods for solving two-step equations, evaluate $f(3)$.

7 [Foundational] • Target: Methods for solving two-step equations

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Methods for solving two-step equations

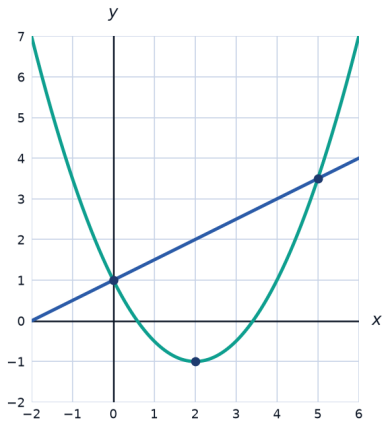
For the model $f(x) = 3x + 3$ in methods for solving two-step equations, evaluate $f(4)$.

9 [Intermediate] • Target: Methods for solving two-step equations

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

10 [Intermediate] • Target: Methods for solving two-step equations

For the model $f(x) = 3x + 3$ in methods for solving two-step equations, evaluate $f(5)$.



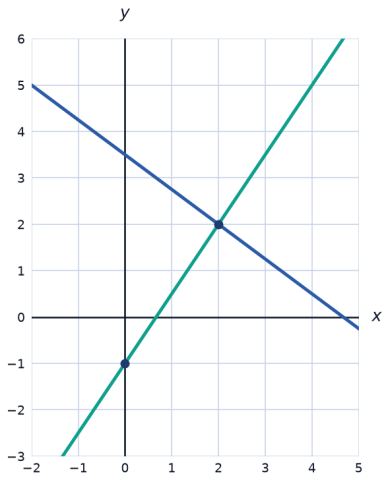
11 [Intermediate] • Target: Methods for solving two-step equations

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

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12 [Intermediate] • Target: Two-step equations

For the model $f(x) = 4x + 4$ in two-step equations, evaluate $f(2)$.



13 [Intermediate] • Target: Two-step equations

A data set recorded for two-step equations consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean? Show your complete algebraic work.

14 [Intermediate] • Target: Two-step equations

For the model $f(x) = 4x + 4$ in two-step equations, evaluate $f(3)$.

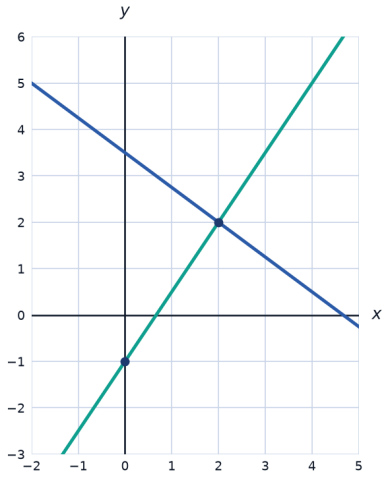
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Two-step equations • Cross-Lesson: 3.01 Coordinate Geometry

For the model $f(x) = 4x + 4$ in two-step equations, evaluate $f(4)$.

16 [Challenging] • Target: Two-step equations • Cross-Lesson: 4.01 Mensuration

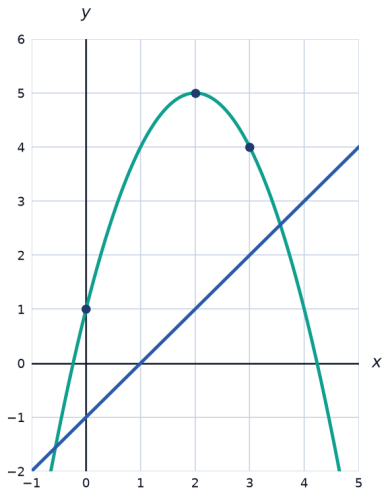
For the model $f(x) = 4x + 4$ in two-step equations, evaluate $f(5)$.



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17 [Challenging] • Target: Solving problems with equations • Cross-Lesson: 5.01 Geometry

For the model $f(x) = 5x + 5$ in solving problems with equations, evaluate $f(2)$.



18 [Challenging] • Target: Solving problems with equations • Cross-Lesson: 6.01 Trigonometry

A data set recorded for solving problems with equations consists of the four values $\{14, 18, 22, 28\}$.

What is the arithmetic mean? Show your complete algebraic work.

19 [Challenging] • Target: Solving problems with equations • Cross-Lesson: 8.01 Probability & Statistics

For the model $f(x) = 5x + 5$ in solving problems with equations, evaluate $f(3)$.

20 [Challenging] • Target: Solving problems with equations • Cross-Lesson: 9.01 Functions

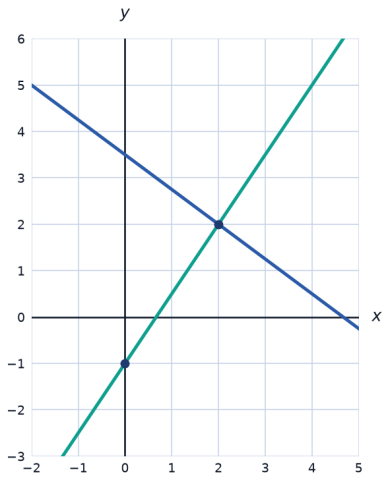
For the model $f(x) = 5x + 5$ in solving problems with equations, evaluate $f(4)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Introduction to equations

For the model $f(x) = 5x + 7$ in introduction to equations, evaluate $f(2)$.



a. 17

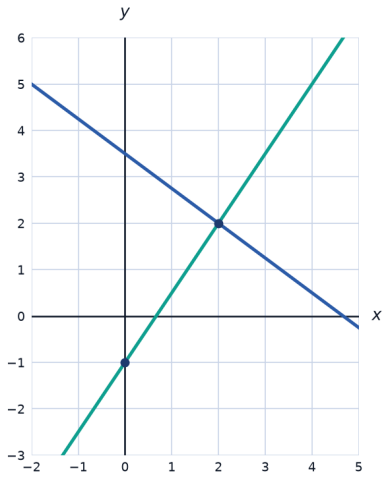
b. 22

c. 10

d. 21

22 [Foundational] • Target: Introduction to equations

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

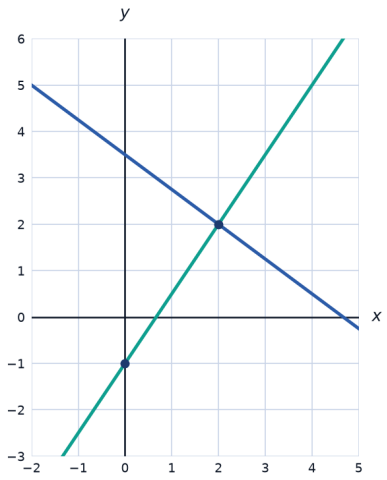
c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

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23 [Foundational] • Target: Introduction to equations

For the model $f(x) = 5x + 7$ in introduction to equations, evaluate $f(5)$.



a. 32

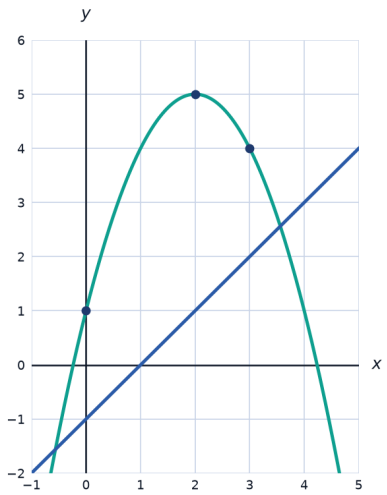
b. 37

c. 25

d. 36

24 [Foundational] • Target: Solutions to equations

For the model $f(x) = 6x + 8$ in solutions to equations, evaluate $f(2)$.



a. 20

b. 26

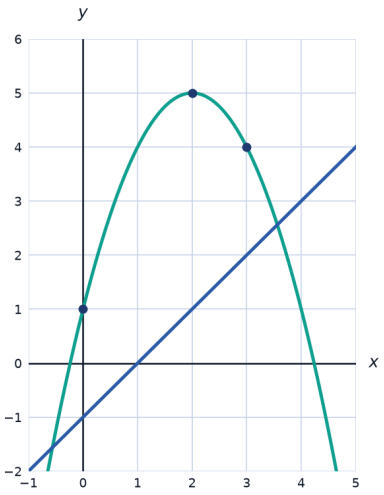
c. 12

d. 24

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25 [Foundational] • Target: Solutions to equations

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

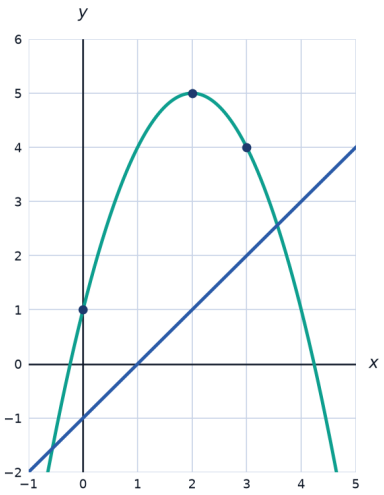
b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

26 [Foundational] • Target: Solutions to equations

For the model $f(x) = 6x + 8$ in solutions to equations, evaluate $f(5)$.



a. 38

b. 44

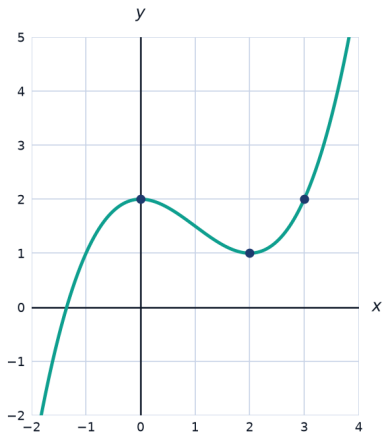
c. 30

d. 42

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27 [Foundational] • Target: Solving equations with algebra

For the model $f(x) = 2x + 9$ in solving equations with algebra, evaluate $f(2)$.



a. 13

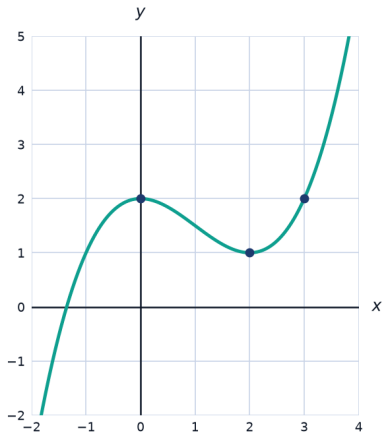
b. 15

c. 4

d. 17

28 [Foundational] • Target: Solving equations with algebra

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Introduction to equations

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ and $x = 4$

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

30 [Intermediate] • Target: Introduction to equations

A data set recorded for introduction to equations consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5

d. 26.5

31 [Intermediate] • Target: Introduction to equations

For the model $f(x) = 5x + 7$ in introduction to equations, evaluate $f(3)$.

a. 22

b. 27

c. 15

d. 26

32 [Intermediate] • Target: Introduction to equations

For the model $f(x) = 5x + 7$ in introduction to equations, evaluate $f(4)$.

a. 27

b. 32

c. 20

d. 31

33 [Intermediate] • Target: Introduction to equations

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

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34 [Intermediate] • Target: Introduction to equations

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

35 [Intermediate] • Target: Solutions to equations

A data set recorded for solutions to equations consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

36 [Intermediate] • Target: Solutions to equations

For the model $f(x) = 6x + 8$ in solutions to equations, evaluate $f(3)$.

a. 26

b. 32

c. 18

d. 30

37 [Intermediate] • Target: Solutions to equations

For the model $f(x) = 6x + 8$ in solutions to equations, evaluate $f(4)$.

a. 32

b. 38

c. 24

d. 36

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Solutions to equations • Cross-Lesson: 6.01 Trigonometry

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

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39 [Challenging] • Target: Solutions to equations • Cross-Lesson: 8.01 Probability & Statistics

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ and $x = 8$

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

40 [Challenging] • Target: Solving equations with algebra • Cross-Lesson: 9.01 Functions

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

41 [Challenging] • Target: Solving equations with algebra • Cross-Lesson: 1.01 Number

For the model $f(x) = 2x + 9$ in solving equations with algebra, evaluate $f(3)$.

a. 15

b. 17

c. 6

d. 19

42 [Challenging] • Target: Solving equations with algebra • Cross-Lesson: 10.01 Sets & Venn Diagrams

For the model $f(x) = 2x + 9$ in solving equations with algebra, evaluate $f(4)$.

a. 17

b. 19

c. 8

d. 21

43 [Challenging] • Target: Solving equations with algebra • Cross-Lesson: 2.01 Algebra & Graphs

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

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44 [Challenging] • Target: Methods for solving two-step equations • Cross-Lesson: 3.01 Coordinate Geometry

A data set recorded for methods for solving two-step equations consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

45 [Challenging] • Target: Two-step equations • Cross-Lesson: 4.01 Mensuration

A data set recorded for two-step equations consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

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